Extending matched SHD training to eighty epochs

exp069 · 19 July 2026 · Draft

Abstract

This one-seed exploratory experiment changes only matched training duration from forty to eighty epochs under the `exp068` recipe. The official SHD test remains sealed: all outcomes use the identical deterministic 7340 training / 816 validation development split. PING selects 48.04% validation accuracy, a 4.54-percentage-point gain over its registered 43.50% baseline. This **registered longer-training signal** and meets the predeclared +3 point threshold. The result supports longer training for this recipe, not a population-level architecture claim.

Methods

The intervention is training duration alone

Both cells use seed 42, 256 excitatory and 64 inhibitory cells, fixed Dale-constrained recurrence, input-weight mean 0.9, 95% input sparsity, a membrane-mean readout scaled by 225, Adam learning rate 0.0004, batch size 32, a 1 ms timestep, and 1000 ms utterance duration. COBA disables the inhibitory loop and uses no voltage-gradient dampening; PING enables the loop and uses dampening 1000. These are the unchanged exp068 settings. Each cell trains for 80 epochs instead of forty.

The official SHD training file is split jointly by speaker and class into 7340 training and 816 validation utterances. Both cells use identical index hashes, which also match exp068. The official SHD test is neither staged nor loaded. Checkpoints maximise validation accuracy, breaking ties by lower validation cross-entropy and then earlier epoch.

Results

Longer training clears the registered PING threshold

PING selects epoch 79 at 48.04% accuracy and 2.029 cross-entropy. Its gain from the exp068 baseline is 4.54 points, exceeding the registered +3 point criterion. COBA selects epoch 75 at 43.87%, a 3.06-point change from its registered 40.81% baseline. The contemporaneous PING–COBA gap is 4.17 points.

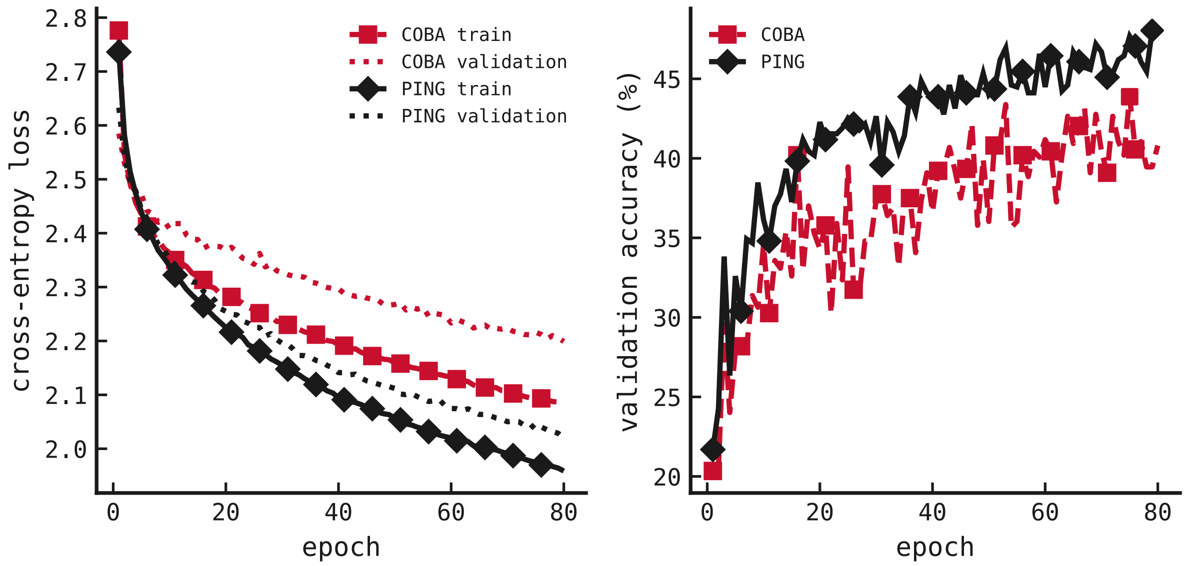

Figure 1: Matched learning curves for the fixed eighty-epoch protocol. Red dashed traces and squares denote COBA; black solid traces and diamonds denote PING. Markers identify checkpoints selected solely from validation accuracy and the registered tie-breaks.

Both selected networks remain active

At its selected checkpoint, COBA's validation excitatory rate is 28.34 Hz. PING's selected validation E/I rates are 10.61 / 49.03 Hz. Neither cell records a skipped update or non-finite forward batch.

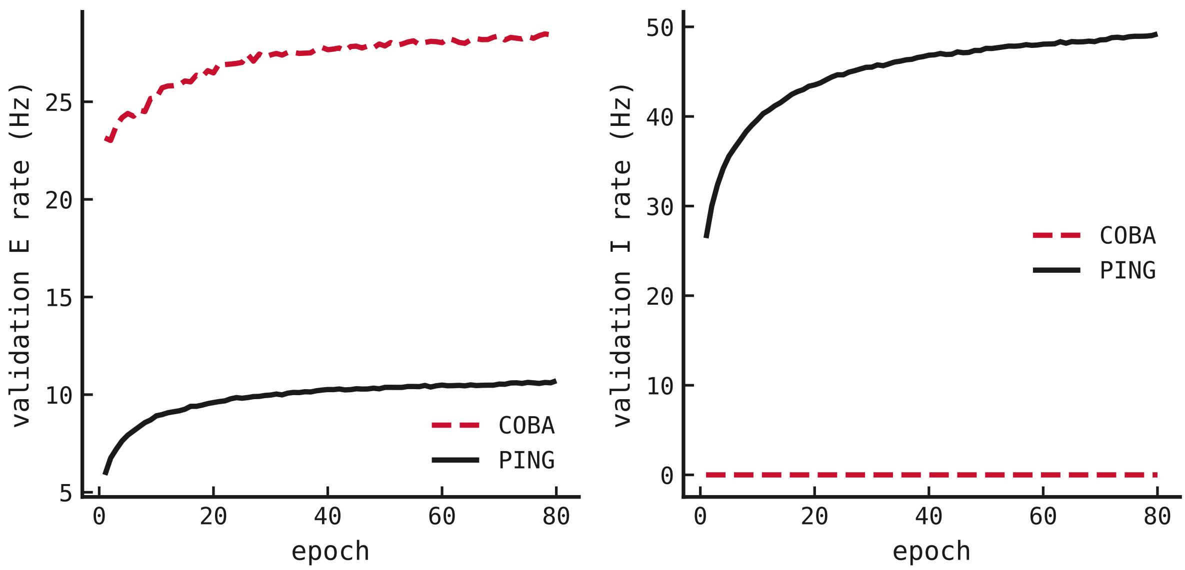

Figure 2: Validation population activity across training. COBA's disabled inhibitory loop remains at zero; PING sustains active excitatory and inhibitory populations.

Matched validation utterances expose different dynamics

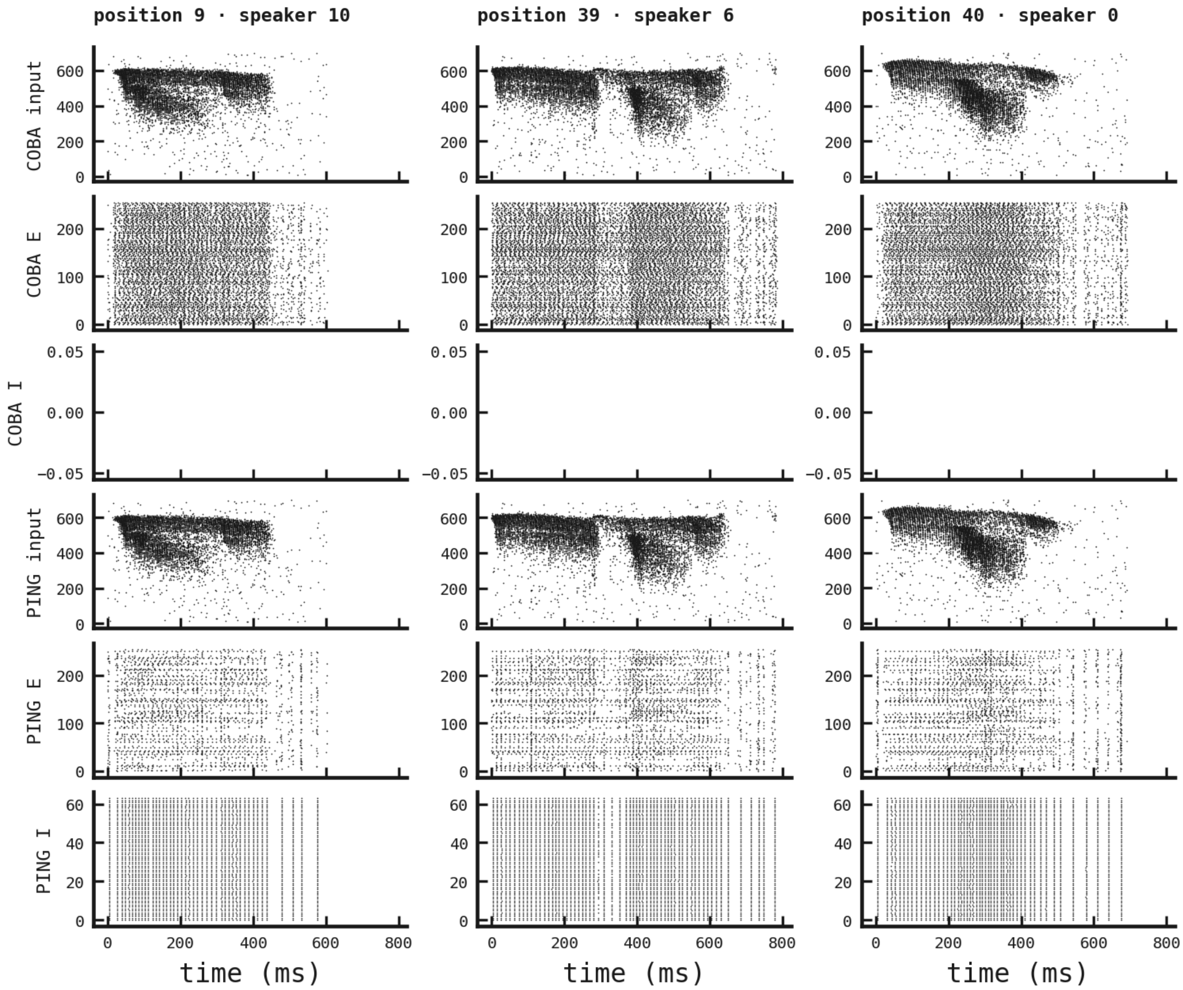

Figure 3: Input, E, and I spike rasters for fixed validation positions 9, 39, 40. Each column uses byte-identical binned input in both cells, with the same time axis and population order.

Verdict and limits

The registered primary criterion is met: extending the unchanged recipe to eighty epochs raises PING validation accuracy by 4.54 points, beyond the required +3 points. PING also remains 4.17 points ahead of contemporaneous COBA. Both selected epochs occur late, so the learning curves—not another unregistered extension—are the appropriate evidence for judging remaining headroom.

This is one seed on one development split. It establishes a useful exploratory training-duration signal but supplies no uncertainty interval and does not justify a general architecture claim. The official SHD test was not accessed. Exact RunPod spend was 4.628 USD, with 0 active pods after collection.